

MPLADS AI INSIGHT & SENTINEL WORKFLOW

AI-Powered System to Detect Anomalies, Fraud & Inefficiencies in MPLADS Implementation

Official Dataset: 543 Lok Sabha MPs (18th House)	Total Tracked Funds: ₹8,320.91 Crore	Repository: github.com/kunalbhalerao19/SIH11
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1. Executive Summary & Problem Context

The **Members of Parliament Local Area Development Scheme (MPLADS)** allocates ₹5 Crore per MP annually (totaling ~₹8,300+ Cr) to fund durable local community infrastructure. However, manual monitoring creates severe vulnerabilities: (1) funds disbursed without corresponding ground progress, (2) double funding of the same asset across central schemes (e.g. PMGSY, Jal Jeevan Mission), (3) photographic fraud and fake geotags, and (4) split-tendering to favor local contractor monopolies. **MPLADS AI Insight** delivers a comprehensive, government-grade oversight platform combining multi-modal AI, Computer Vision, Geospatial Intelligence, OCR document auditing, and citizen verification into an explainable, automated fraud detection sentinel.

2. Official 18th Lok Sabha Master Dataset

Metric	Official Value	Scope & Operational Significance
Total MPs Monitored	543 Members	100% of 18th Lok Sabha across all 36 States & Union Territories
Total Tracked Funds	₹8,320.91 Crore	Authentic allocation sums from the official MPLADS portal records
Total Expenditure	₹6,740.07 Crore	Nationwide fund utilization (~81.0% overall expenditure rate)
Works Monitored	67,982 Recommended	44,873 Completed, 11,480 Ongoing, 1,342 High-Risk Flags
State Coverage	36 States & UTs	Real-time aggregated performance grades (A/B/C/D) and utilization rates

3. Dedicated Multi-Modal AI / Analytics Sub-Modules

The architecture implements 6 specialized sub-engines, each evaluating distinct fraud vectors and outputting an independent Sub-Risk Score (0–100):

AI Sub-Module	Weight	Key Features & Algorithmic Detection Signals	Sub-Score
1. Financial Anomaly	25%	Benford's Law distribution analysis, 'March Rush' velocity bursts (year-end spending), cost overrun deviations vs. sector averages.	0–100
2. Photo & CV Evidence	20%	Perceptual Hashing (pHash) for duplicate/stock photo reuse, EXIF GPS geotag distance delta vs. site (>100m flag), CNN visual progress estimation.	0–100
3. Geospatial Intelligence	15%	Boundary coordinate validation, spatial density clustering (DBSCAN), duplicate asset proximity search (<100m buffer zone).	0–100
4. Vendor & Cartels	15%	Herfindahl-Hirschman Index (HHI > 2500 monopoly flag), split-tendering pattern detection, price benchmarking vs. CPWD Schedule of Rates.	0–100

5. Document Intelligence	10%	OCR & text extraction from DPRs, Sanction Orders & UCs; invoice-to-sanction reconciliation; pre-dated UC temporal impossibility detection.	0–100
6. Progress & Timeline	15%	S-Curve milestone tracking, progress-expenditure gap divergence (e.g. 90% funds spent vs 18% physical progress), critical path delay velocity.	0–100

4. Evidence Fusion & Cross-Scheme Duplication Engine

Multi-Modal Evidence Fusion: All 6 sub-scores are synthesized via a calibrated weighted fusion function: *Composite Score* = $\sum (w_i \times SubScore_i) / \sum w_i$. The system categorizes works into **LOW (0–30)**, **MEDIUM (31–60)**, **HIGH (61–80)**, and **CRITICAL (81–100)** with an Explainable AI (XAI) waterfall breakdown showing exact point attribution per domain.

Cross-Scheme Duplication Engine: Automatically cross-references MPLADS assets against other major central/state infrastructure schemes (**PMGSY, Jal Jeevan Mission, AMRUT, Smart Cities Mission**) using coordinate proximity, semantic NLP title matching, and fiscal timeline overlaps to detect and prevent double funding of the exact same asset.

MPLADS Project	Overlapping Scheme	Proximity	Title Match	Double-Funding Exposure
MPL-2026-00451 (Hadapsar Community Hall)	Smart Cities (SCM-MH-PUN-0842)	48 meters	87% Match	■85 Lakhs (CRITICAL)
MPL-2026-00102 (Phulwari Road Link)	PMGSY (PMGSY-BR-PAT-4401)	55 meters	91% Match	■120 Lakhs (CRITICAL)
MPL-2026-04131 (Begusarai Piped Water)	Jal Jeevan (JJM-BR-BEG-2918)	32 meters	82% Match	■180 Lakhs (CRITICAL)

5. Comprehensive Application Page & Feature Catalog

Module / Route	Primary Purpose & Interactive Functionality
1. Landing Page (/)	Official MoSPI portal landing with nationwide metrics, workflow summary, and public access points.
2. Dashboard (/dashboard)	8 primary executive KPIs, 5 Recharts visualizations (fund vs util, status donut, risk trend, sector spend) and top suspicious works.
3. AI Analytics Lab (/ai-analytics)	Multi-Modal Sentinel workspace: 6 sub-module diagnostic bench, dynamic weight sliders, radar chart, and cross-scheme detector.
4. AI Risk Analysis (/ai-risk)	Forensic project-level explainable AI (XAI) breakdown with SHAP feature bars, peer project comparator, and timeline audit.
5. Anomaly Center (/anomalies)	Master ledger of 287+ AI-detected anomalies filterable by severity, type, state, and officer review status.
6. Projects Directory (/projects)	Complete searchable database of all works with multi-criteria filters (FY, State, House, Sector, Status, Risk).
7. Project Forensic (/projects/:id)	Detailed investigation dossier with financial progress bars, stage-gate tracking, document & photo verification placeholders.
8. Fund Monitoring (/fund-monitoring)	Multi-year financial release/utilization tracking, state comparison charts, and paginated 543 MP-wise fund utilization table.
9. MP Analytics (/mp-analytics)	543 Lok Sabha MP Explorer: Search by name/constituency, MP profile card, 3 AI efficiency gauges, and 543 MP master table.
10. State Analytics (/state-analytics)	Performance grading (A/B/C/D) across 36 States/UTs, completion bar charts, and risk heatmaps.
11. Map View (/map)	Interactive India state grid matrix color-coded by Expenditure, Completion %, Risk, Anomalies, or Total Works.
12. Alert Center (/alerts)	Triage alert queue with operational review actions: Assign Officer, Record Investigation Comments, and Update Status.
13. Reports Dossier (/reports)	9 automated report types (Anomaly Summary, State Performance, MP Utilization, Delayed Works) with 1-click PDF/CSV export.
14. Data Explorer (/data-explorer)	Advanced MIS multi-query builder for auditors with column sorting, live record counts, and full data export.
15. AI Pipeline (/ai-model)	Architecture map of ML models (Isolation Forest, XGBoost, K-Means, ARIMA), feature weights, and simulated precision/recall.
16. Citizen Portal (/citizen)	Public transparency portal for local constituency search and grievance reporting without exposing internal AI risk scores.

6. Inspection Priority, Administrative Triage & Feedback Loop

Automated Inspection Priority Queue: AI dynamically ranks suspicious projects by risk severity and total public funds at risk. Vigilance officers can generate a standardized **1-Click Physical Inspection Docket** equipped with on-site GPS verification coordinates, material test requirements, and photographic angle guidelines for field inspectors.

Citizen QR Verification & Continuous Learning Loop: Each sanctioned project generates a unique QR code posted on site signboards. Citizens scan the QR to submit geo-tagged field observations and photos. Validated citizen reports directly update the machine learning model's feature weights and confidence parameters in an active feedback loop (Step 10).

7. Technical Specifications & Government Design System

Architecture Layer	Technology & Compliance Standard
Frontend Framework	React 19, TypeScript 6.0, React Router v7
Styling & Tokens	Tailwind CSS v4 with Government of India Design System tokens (#003580 primary)
Data Visualizations	Recharts v3 (Responsive Bar, Line, Donut, Radar, and Heatmap components)
Build & Bundling	Vite 8 with Rolldown compiler (sub-second HMR, optimized production chunks)
Legal & AI Phrasing	Objective neutral framing ('Potential anomaly detected — requires official verification')

SMART INDIA HACKATHON 2026 PROTOTYPE SUMMARY
GitHub Repository: <https://github.com/kunalbhalerao19/SIH11>
Live Dev URL: <http://localhost:5173/>
Developed for Problem Statement 26102 (Ministry of Statistics and Programme Implementation).